Physics of rubber — Solution

Y26 (2026) — English translation

A1 — 0.40 pts

Find the number of microscopic states Γ, with the help of which the macroscopic state is realized, in which the difference in the number of links oriented along and against the axis x is equal to $m = n_1 - n_2$. Also obtain an approximate answer for $m \ll n$. Express your answers using m and n.

For a given m, the number of links oriented in the positive direction is $n_1 = \dfrac{n+m}{2}$, and in the negative direction $n_2 = \dfrac{n-m}{2}$. Then the required number of ways

Answer:

$$\Gamma = \frac{n!}{\left(\frac{n+m}{2}\right)!\left(\frac{n-m}{2}\right)!} \approx 2^n \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{m^2}{2n}\right)$$

A2 — 0.80 pts

Show that at $h_x \ll na$ the entropy of the molecule has the form

$$S = S_0 - \gamma h_x^2.$$

Express coefficient γ through n, a and k.

The displacement of the end of the molecule is $h_x = (n_1 - n_2)a = ma$. Then, using the formula from the previous paragraph, we get

$$\Gamma = 2^n \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{h_x^2}{2na^2}\right).$$

Then from the definition entropy obtain

$$S = k \ln \Gamma = k \ln \left(2^n \sqrt{\frac{2}{\pi n}}\right) - \frac{k h_x^2}{2na^2}.$$

Answer:

$$\gamma = \frac{k}{2na^2}$$

A3 — 0.60 pts

The probability that the end of a molecule lies in the interval $[h_x, h_x + dh_x]$ of width $dh_x \gg a$ is $f(h_x)dh_x$. Show that for $|h_x| \ll na$ the probability density of $f(h_x)$ has the form

$$f(h_x) = A \exp(-\alpha h_x^2)$$

and express constant A and α through n and a. Note, that direction element/link chain are independent of each other and are equally probable. Note: for rotation one element/link h_x change on $2a$.

Note: When one link is rotated, h_x changes to $2a$.

The total number of states in the circuit is 2^n. Since the directions of the chain links are equally probable, the probability of a given state of the molecule is proportional to the number of microscopic states in which it can be realized. Then the probability that the displacement of the end of the molecule takes on a given value h_x is equal to

$$p(h_x) = \frac{1}{2^n}\Gamma(h_x) = \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{h_x^2}{2na^2}\right).$$

For rotation one element/link value m change on 2, therefore displacement h_x change on $2a$. Means in some interval width $dh_x \gg a$ be located $dN = dh_x/2a$ possible state. For sufficiently small dh_x probability this state is practically the same, and means probability of falling within the given interval equal

$$dN \cdot p(h_x) = \frac{dh_x}{2a}\sqrt{\frac{2}{\pi n}} \exp\left(-\frac{h_x^2}{2na^2}\right).$$

Hence density probability

$$f(h_x) = \frac{1}{a\sqrt{2\pi n}} \exp\left(-\frac{h_x^2}{2na^2}\right).$$

Taking into account expression for Gaussian integral

$$\int\limits_{-\infty}^{+\infty} dx\, e^{-\beta x^2} = \sqrt{\frac{\pi}{\beta}}$$

obtain

$$\int\limits_{-\infty}^{+\infty} dh_x\, f(h_x) = \frac{1}{a\sqrt{2\pi n}}\sqrt{2\pi n a^2} = 1,$$

that is, the distribution is correctly normalized.

Answer:

$$A = \frac{1}{a\sqrt{2\pi n}}, \quad \alpha = \frac{1}{2na^2}$$

B1 0.50 pts

What are the values of $\langle h^2 \rangle$ and $\langle h_x^2 \rangle$ for a three-dimensional chain? Express your answers using n and l. Note: if you were unable to complete this step, you can continue to count $\langle h_x^2 \rangle = nl^2$ throughout the entire task.

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The displacement vector of the end of the vector relative to the beginning is equal to the sum of the vectors $\vec{l}_i$,

$$\vec{h} = \sum_i \vec{l}_i,$$

therefore it square

$$\vec{h}^2 = \left(\sum_i \vec{l}_i\right)^2 = \sum_i (\vec{l}_i)^2 + 2\sum_{i<j} \vec{l}_i \vec{l}_j.$$

Here in first sum the summation is over all element/link molecule, and in second – by all pair element/link. Each term first sum equal length one element/link l^2, therefore all first sum equal nl^2. Average second sum by various state molecule. due to the independence of direction various element/link each from product for averaging give 0, $\langle \vec{l}_i \vec{l}_j \rangle = 0$, therefore and all sum for averaging becomes in 0. Then for mean square obtain

$$\langle h^2 \rangle = nl^2.$$

in force equal probability all direction

$$\langle h_x^2 \rangle = \langle h_y^2 \rangle = \langle h_z^2 \rangle.$$

Since

$$\langle h^2 \rangle = \langle h_x^2 \rangle + \langle h_y^2 \rangle + \langle h_z^2 \rangle,$$

mean square one from projection equal

$$\langle h_x^2 \rangle = \frac{1}{3}\langle h^2 \rangle = \frac{1}{3} n l^2.$$

Answer:

$$\langle h^2 \rangle = n l^2, \quad \langle h_x^2 \rangle = \frac{1}{3} n l^2$$

B2 — 1.00 pts

What is the average distance between the ends of the chain $\langle h \rangle$? Express your answer in terms of n and l.

Differentiating the value of the Gaussian integral

$$\int_{-\infty}^{+\infty} dx\, e^{-\beta x^2} = \sqrt{\frac{\pi}{\beta}}$$

by parameter β obtain

$$\int_{-\infty}^{+\infty} dx\, x^2 e^{-\beta x^2} = \frac{1}{2}\sqrt{\frac{\pi}{\beta^3}}, \quad \int_{-\infty}^{+\infty} dx\, x^4 e^{-\beta x^2} = \frac{3}{4}\sqrt{\frac{\pi}{\beta^5}}.$$

From the condition normalization density probability

$$\int d^3 h f(\vec{h}) = 1 = A \int d^3 h e^{-\beta h^2} = 4\pi A \int_0^{\infty} h^2 dh\, e^{-\beta h^2} = \pi A \frac{\sqrt{\pi}}{\beta^{3/2}}$$

find coefficient

$$A = \left(\frac{\beta}{\pi}\right)^{3/2}.$$

Note, that for calculation integral appears additional factor 1/2, since lower limit equal 0, and not $-\infty$. Then for mean square obtain

$$\langle h^2 \rangle = \int d^3 h\, h^2 f(\vec{h}) = 4\pi A \int_0^{\infty} h^4 dh\, e^{-\beta h^2} = \pi A \frac{3}{2} \frac{\sqrt{\pi}}{\beta^{5/2}} = \frac{3}{2\beta}.$$

Use expression from previous part, find

$$\frac{3}{2\beta} = n l^2,$$

from which

$$\beta = \frac{3}{2 n l^2}.$$

Let us compute mean magnitude/modulus displacement

$$\langle h \rangle = \int d^3 h\, h f(\vec{h}) = 4\pi A \int_0^{\infty} h^3 dh\, e^{-\beta h^2}.$$

Transition to variable $\xi = h^2$:

$$\langle h \rangle = 2\pi A \int_0^{\infty} \xi d\xi\, e^{-\beta\xi} = 2\pi A \left(-\frac{1}{\beta} \xi e^{-\beta\xi} \Big|_0^{\infty} + \frac{1}{\beta} \int_0^{\infty} d\xi\, e^{-\beta\xi} \right) = \frac{2\pi A}{\beta^2} = \frac{2}{\sqrt{\pi\beta}}.$$

Substitute value β, we finally find

The same result can be obtained using an analogy with the Maxwell distribution. Indeed, the distribution of the displacement vector of the second vector of the end of the molecule relative to the first differs from the distribution of the velocity vector only in the value of the parameter β. For the Maxwell distribution, the root mean square value of the velocity

$$\langle v^2 \rangle = \frac{3kT}{m},$$

and also value mean magnitude/modulus velocity

$$\langle v \rangle = \sqrt{\frac{8kT}{\pi m}}.$$

Hence we get dimensionless ratio

$$\frac{\langle v \rangle^2}{\langle v^2 \rangle} = \frac{8}{3\pi}.$$

It not contains parameter distribution, therefore such same relation must hold and for $\vec{h}$:

$$\frac{\langle h \rangle^2}{\langle h^2 \rangle} = \frac{8}{3\pi},$$

is known, from which we immediately obtain the above answer.

Answer:

$$\langle h \rangle = \sqrt{\frac{8n}{3\pi}} l$$

B3 1.50 pts

Find the average displacement of the second end of the molecule h. Express your answer in terms of F, T, n, l and k. Obtain an exact expression and an approximate one for small F.

From symmetry, the average displacement vector of the end of the molecule will be directed along the external force, so you only need to find the projection of this vector onto the direction of the force. Let's consider some link, let it form an angle θ with the direction of the force. Then its potential energy $U = -Fl\cos\theta$, which means the probability that it is directed at some solid angle $d\Omega = 2\pi\sin\theta d\theta$ depends only on θ and has the form

$$B\exp\left(-\frac{U}{kT}\right) d\Omega = B\exp\left(\frac{Fl}{kT}\cos\theta\right) d\Omega,$$

where the coefficient B can be found from normalization. For convenience of calculation, we introduce the dimensionless variable $\xi = \dfrac{Fl}{kT}$. Then

$$B\int_0^\pi e^{\xi\cos\theta} 2\pi\sin\theta d\theta = 1.$$

Going over to the variable of integration $u = \cos\theta$, which varies from -1 to 1, we obtain

$$2\pi B\int_{-1}^{1} e^{\xi u} du = \frac{2\pi B}{\xi}(e^{\xi} - e^{-\xi}) = \frac{4\pi B}{\xi}\sinh\xi = 1,$$

from which

$$B = \frac{1}{4\pi}\frac{\xi}{\sinh\xi}.$$

Then the mean projection of a link on the direction of the force is

$$\langle l\cos\theta \rangle = B\int_0^\pi l\cos\theta e^{\xi\cos\theta} 2\pi\sin\theta d\theta = 2\pi Bl\int_{-1}^{1} u\, e^{\xi u}\, du.$$

Compute this integral by parts:

$$\int_{-1}^{1} u\, e^{\xi u}\, du = \frac{1}{\xi}\, u\, e^{\xi u}\Big|_{-1}^{1} - \frac{1}{\xi}\int_{-1}^{1} e^{\xi u}\, du = \frac{1}{\xi}(e^{\xi} + e^{-\xi}) - \frac{1}{\xi^2}(e^{\xi} - e^{-\xi}).$$

$$\langle l\cos\theta\rangle = \frac{l}{2}\frac{\xi}{\sinh\xi} 2\left(\frac{\cosh\xi}{\xi} - \frac{\sinh\xi}{\xi^2}\right) = l\left(\coth\xi - \frac{1}{\xi}\right).$$

The total displacement of the end of the molecule consists of the projections of the individual links:

$$h = n\langle l\cos\theta\rangle = nl\left(\coth\frac{Fl}{kT} - \frac{kT}{Fl}\right).$$

In the limit small force obtain expansion

$$\coth\xi - \frac{1}{\xi} = \frac{e^{\xi}+e^{-\xi}}{e^{\xi}-e^{-\xi}} - \frac{1}{\xi} \approx \frac{2+\xi^2}{2\xi+\xi^3/3} - \frac{1}{\xi} = \frac{1}{\xi}\left(\frac{1+\xi^2/2}{1+\xi^2/6} - 1\right) = \frac{1}{\xi}\frac{\xi^2}{3} = \frac{\xi}{3}.$$

Then for displacement obtain

$$h \approx \frac{nl}{3}\frac{Fl}{kT} = \frac{nl^2}{3kT}F.$$

Note also that in the limit of large force, i.e. $\xi \to \infty$, we obtain $\coth\xi \to 1$ and

$$h \approx nl\left(1 - \frac{kT}{Fl}\right) \approx nl,$$

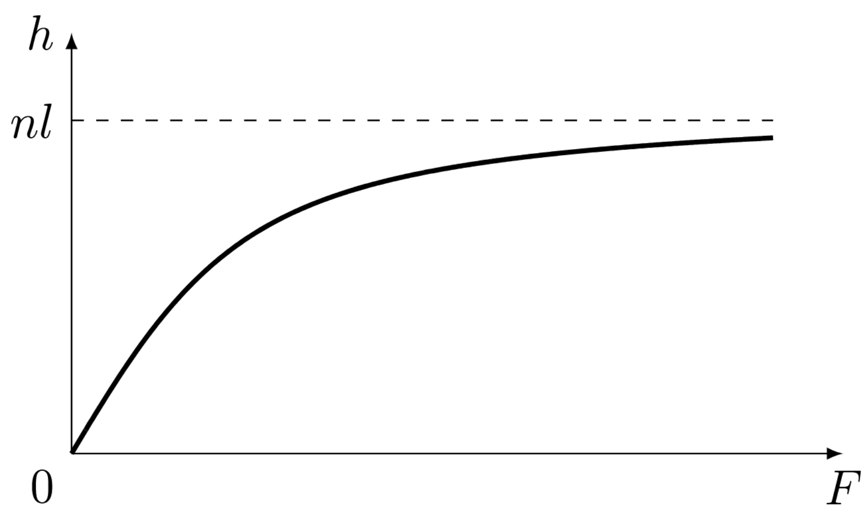
that is the molecule is practically completely straightened. The characteristic value of the force at which the transition from the small-force regime to the large-force regime occurs corresponds to $\xi \sim 1$, that is $F \sim kT/l$.

Answer:

$$h = nl\left(\coth\frac{Fl}{kT} - \frac{kT}{Fl}\right) \approx \frac{nl^2}{3kT}F$$

B4 — 0.50 pts

Plot a graph of the length of the molecule h versus the applied force. Provide typical values.



C1 — 0.70 pts

What is the change in entropy ΔS of the entire mesh as a result of stretching? Express your answer in terms of k, λ_x, λ_y, λ_z and N. Note: if you were unable to make B1, then use $\langle h_x^2\rangle = nl^2$.

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The entropy of one chain in a stretched state is equal to

$$S_1(\vec{h'}) = S_0 - \frac{3k}{2nl^2}(\lambda_x^2 h_x^2 + \lambda_y^2 h_y^2 + \lambda_z^2 h_z^2).$$

The change in the entropy of one chain as a result of stretching

$$\Delta S_1 = S_1(\vec{h'}) - S_1(\vec{h}) = -\frac{3k}{2nl^2}((\lambda_x^2 - 1)h_x^2 + (\lambda_y^2 - 1)h_y^2 + (\lambda_z^2 - 1)h_z^2).$$

The change in the entropy of the polymer network is expressed through the average value of the change in the entropy of one chain:

$$\Delta S = N\langle \Delta S_1\rangle = -\frac{3kN}{2nl^2}((\lambda_x^2 - 1)\langle h_x^2\rangle + (\lambda_y^2 - 1)\langle h_y^2\rangle + (\lambda_z^2 - 1)\langle h_z^2\rangle).$$

Using the result of point B, we find

Answer:

$$\Delta S = -\frac{kN}{2}(\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)$$

C2 — 0.60 pts

Let us consider tension in the case when forces are applied along one of the axes and stretch the sample along this axis by a factor of λ, and there are no stresses along the other axes. What is the change in entropy of the sample? Express your answer in terms of k, λ and N.

If forces are applied only along the x axis, then $\lambda_x = \lambda$, and for other axes $\lambda_y = \lambda_z$ is true. Since the volume of the rod does not change, then $\lambda_x\lambda_y\lambda_z = 1$. Then $\lambda_y = \lambda_z = 1/\sqrt{\lambda}$. Substituting the dilations into the formula from C2, we find

Answer:

$$\Delta S = -\frac{kN}{2}\left(\lambda^2 + \frac{2}{\lambda} - 3\right)$$

C3 — 0.60 pts

Using the first law of thermodynamics for an isothermal process and the result of part C2, show that the equation of state of rubber (the dependence of the elastic force on tension and temperature) is:

$$F(\lambda, T) = \beta(T)\left(\lambda - \frac{1}{\lambda^2}\right).$$

Find β. Express your answer in terms of k, N, T and the length of the sample in the unstretched state l_0.

The first beginning in general form is written as

$$dU = \delta Q + \delta A.$$

Here the supplied heat one can express through change entropy: $\delta Q = TdS$. Under work δA for such choice sign is meant the work, performed on sample. It in general case consists of from work force pressure and work external force, stretch rubber

$$\delta A = F dl - p\,dV.$$

Then we finally obtain

$$dU = T\,dS + F\,dl - p\,dV,$$

where p is the external pressure and dV is the change in volume of the rod. Since the volume of the rod does not change, the external pressure does no work. Since the internal energy depends only on temperature, in an isothermal process $dU = 0$. Then

$$F = -T\left(\frac{\partial S}{\partial l}\right)_T = -\frac{T}{l_0}\left(\frac{\partial \Delta S}{\partial \lambda}\right)_T = \frac{kNT}{l_0}\left(\lambda - \frac{1}{\lambda^2}\right).$$

Answer:

$$\beta = \frac{kNT}{l_0}$$

C4 — 0.80 pts

Show that Hooke's law is valid for small deformations. Find the Young's modulus E of the rubber. Express your answer in terms of k, T and the concentration ν of polymer chains in the sample. Calculate the numerical value of Young's modulus if the rubber density is $\rho = 1200\ kg/m^3$, molar mass $\mu = 10^5\ g/mol$, temperature $T = 300\ K$.

Let us substitute into the expression for force $\lambda = 1 + \varepsilon$ at $\varepsilon \ll 1$:

$$F = \frac{kNT}{l_0}\left(1 + \varepsilon - \frac{1}{(1+\varepsilon)^2}\right) \approx \frac{3kNT\varepsilon}{l_0}.$$

Thus, for small deformations, the force is directly proportional to the relative elongation, i.e. Hooke's law is true. Young's modulus is equal to (S_0 – initial cross-sectional area)

$$E = \frac{F}{S_0\varepsilon} = \frac{3kNT}{l_0 S_0} = \frac{3kNT}{V} = 3k\nu T.$$

The concentration of polymer chains can be determined by knowing the rubber density and molar mass:

$$\nu = \frac{\rho}{\mu} N_A,$$

since ρ/μ – the number of moles per unit volume. Taking into account the numerical values

$$\nu = \frac{\Delta N}{\Delta V} = \frac{N_A \Delta m/\mu}{\Delta V} = \frac{\rho N_A}{\mu} = 7.2 \cdot 10^{24}\ \mathrm{m}^{-3},\ E \approx 9 \cdot 10^4\ \mathrm{Pa}.$$

Answer:

$$E = 3k\nu T \approx 9 \cdot 10^4\ \mathrm{Pa}$$

D1 — 0.20 pts

Write down the expression for U_{therm}. Express your answer in terms of the heat capacity at constant length C_l and T. Consider that C_l does not depend on temperature.

Answer:

$$U_{\mathrm{therm}} = C_l T$$

D2 — 1.20 pts

Find an expression for U_{mech}. Express your answer in terms of V, ε, T, $E(T)$ and $dE(T)/dT$.

As follows from the expression for entropy, during isothermal tension the entropy of the rod decreases. This means that heat is supplied on an isotherm with a higher temperature, and the Carnot cycle occurs in the forward direction. On an isotherm with a higher temperature, the rod is compressed from ε to zero deformation with a constant Young's modulus $E = E(T + dT)$ and does positive work

$$A_+ = \int\limits_0^{\varepsilon} E\varepsilon S_0 l_0\, d\varepsilon = \frac{E\varepsilon^2 V}{2}.$$

The change in internal energy on the isotherm is equal to the change in elastic strain energy, since U_{therm} depends only on temperature:

$$\Delta U_+ = U_{\text{mech}}(0, T + dT) - U_{\text{mech}}(\varepsilon, T + dT) = 0 - U_{\text{mech}}(\varepsilon, T + dT) = -U_{\text{mech}}.$$

The heat received from the heater is

$$Q_+ = A_+ + \Delta U_+ = \frac{E\varepsilon^2 V}{2} - U_{\text{mech}}.$$

By condition work on adiabats can be neglected. Then the work of the rod per cycle is equal to

$$dA = \frac{E(T + dT)\varepsilon^2 V}{2} - \frac{E(T)\varepsilon^2 V}{2} = \frac{\varepsilon^2 V\, dE}{2}.$$

Carnot cycle efficiency

$$\eta = \frac{dT}{T} = \frac{dA}{Q_+}.$$

From here we find

Answer:

$$U_{\text{mech}} = \frac{\varepsilon^2 V}{2}\left(E - T\frac{dE}{dT}\right)$$

.

D3 **0.40 pts**

Let the Young's modulus of rubber is equal to to E_0 at temperature T_0. Find Young's modulus E at temperature T in the "ideal rubber" model. Express your answer using E_0, T_0 and T.

In the "ideal rubber" model $U_{\text{mech}} = 0$. Taking D2 into account, we have

$$E - T\frac{dE}{dT} = 0,$$

from which

$$\ln\frac{E}{E_0} = \ln\frac{T}{T_0}.$$

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where

$$\ln\frac{E}{E_0} = \ln\frac{T}{T_0}.$$

Answer:

$$E = E_0\frac{T}{T_0}$$

E1 **0.80 pts**

Obtain an expression for ΔT. Express your answer in terms of E_0, S_0, l_0, C_l, α, T_0 and λ. Consider that $\Delta T = T - T_0 \ll T_0$.

Let's differentiate the equation of state with respect to temperature at a constant length:

$$\left(\frac{\partial F}{\partial T}\right)_l = \frac{E_0 S_0}{3T_0}\left(\lambda - \frac{1+3\alpha(T-T_0)}{\lambda^2}\right) - \frac{E_0 S_0 T}{3T_0}\cdot\frac{3\alpha}{\lambda^2} = \\ = \frac{E_0 S_0}{3T_0}\left(\lambda - \frac{1+3\alpha(T-T_0)+3\alpha T}{\lambda^2}\right) \approx \frac{E_0 S_0}{3T_0}\left(\lambda - \frac{1+3\alpha T_0}{\lambda^2}\right). \quad (1)$$

The last transition used the condition $\Delta T \ll T_0$. Substituting the resulting expression into the relation for the temperature change, we find

$$\Delta T = \frac{E_0 S_0}{3C_l}\int_1^\lambda \left(\lambda - \frac{1+3\alpha T_0}{\lambda^2}\right) l_0\, d\lambda = \\ = \frac{E_0 S_0 l_0}{3C_l}\left(\frac{\lambda^2-1}{2} + (1+3\alpha T_0)\left(\frac{1}{\lambda}-1\right)\right) = \frac{E_0 S_0 l_0}{3C_l}(\lambda-1)\left(\frac{\lambda+1}{2} - \frac{1+3\alpha T_0}{\lambda}\right). \quad (2)$$

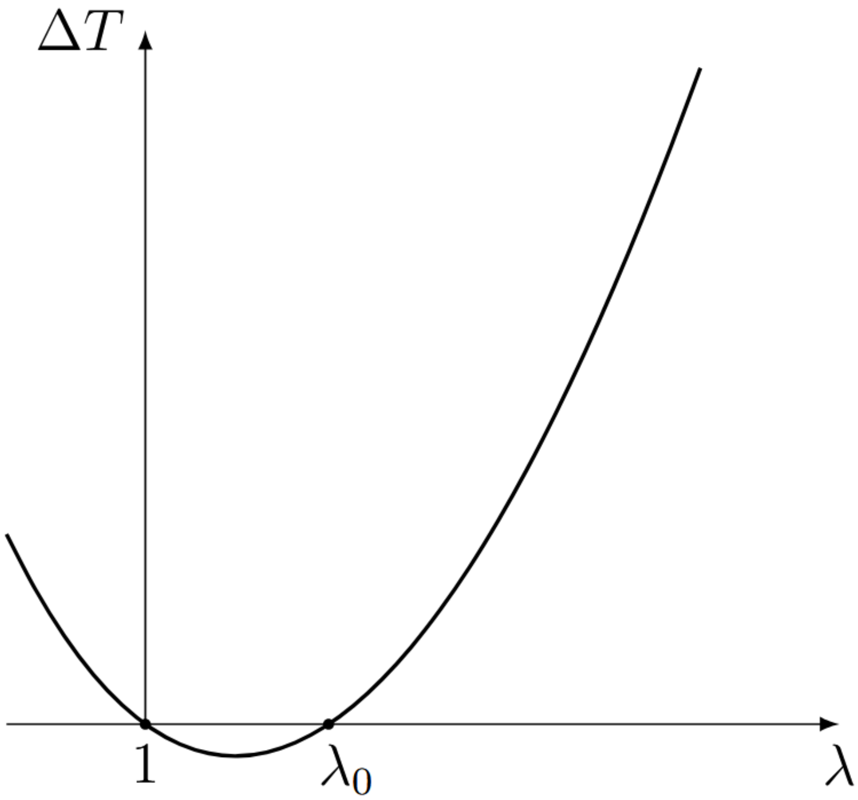
Answer:

$$\Delta T = \frac{E_0 S_0 l_0}{3C_l}(\lambda-1)\left(\frac{\lambda+1}{2} - \frac{1+3\alpha T_0}{\lambda}\right).$$

E2 0.50 pts

Construct a high-quality graph $\Delta T(\lambda)$.

Using the resulting formula, we will construct a qualitative graph $\Delta T(\lambda)$.



E3 0.50 pts

At what stretching λ_0 is the sign change of ΔT observed? Give a numerical value as your answer.

At $\lambda = \lambda_0$ the expression in right parentheses becomes zero:

$$\frac{\lambda+1}{2} - \frac{1+3\alpha T_0}{\lambda} = 0 \Rightarrow \lambda^2 + \lambda - 2(1+3\alpha T_0) = 0.$$

The positive root of this equation is

$$\lambda_0 = \frac{-1+\sqrt{1+8(1+3\alpha T_0)}}{2} \approx 1 + 2\alpha T_0 = 1.132.$$

Answer:

$$\lambda_0 \approx 1.132$$

E4 **0.40 pts**

What is the change in temperature $\Delta T_{1,2}$ of the sample under tension $\lambda_1 = 1.1$ and $\lambda_2 = 1.4$? Use $l_0 = 1$ m and $C_l = 150$ J/K for calculations.

Using the formula from E1, we find the temperature changes:

Answer:

$$\Delta T_1 = -1.05 \cdot 10^{-3} \text{ K}$$

$$\Delta T_2 = 3.00 \cdot 10^{-2} \text{ K}$$